

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA1407

COURSE NAME: Ethical Hacking

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REG NO: 192565025

COURSE OUTCOME COVERED

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

Assessment Tool Weightage: 40%

```
nmap -sn scanme.nmap.org
nmap -sT scanme.nmap.org
nmap -sV scanme.nmap.org
sudo nmap -O scanme.nmap.org
sudo nmap -sC scanme.nmap.org
sudo nmap -A scanme.nmap.org
sudo nmap -p- -sV -O -sC scanme.nmap.org
```

```

lavanya@kali: ~
Session Actions Edit View Help

(lavanya@kali)-[~]
$ nmap -sn scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:12 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

(lavanya@kali)-[~]
$ nmap -sT scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:12 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

(lavanya@kali)-[~]
$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:12 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

(lavanya@kali)-[~]
$ sudo nmap -O scanme.nmap.org
[sudo] password for lavanya:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:13 +0530
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

(lavanya@kali)-[~]
$ sudo nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:13 +0530
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

(lavanya@kali)-[~]
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:14 +0530
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

(lavanya@kali)-[~]
$ sudo nmap -p- -sV -O -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:14 +0530
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done

```

2. Metasploitable 2 - 192.168.56.101

```
nmap -sn 192.168.56.101
```

```
nmap -p- 192.168.56.101
```

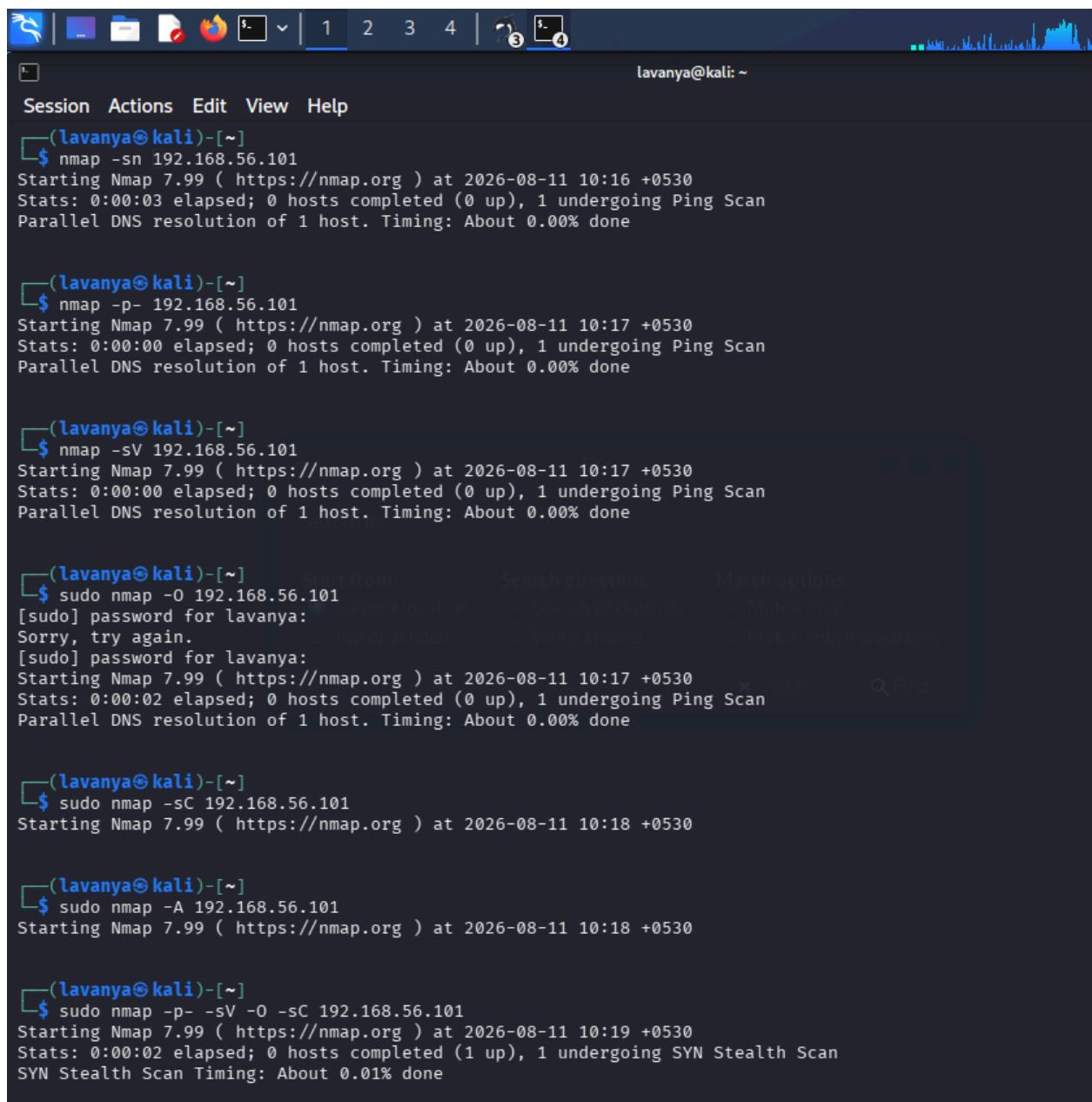
```
nmap -sV 192.168.56.101
```

```
sudo nmap -O 192.168.56.101
```

```
sudo nmap -sC 192.168.56.101
```

```
sudo nmap -A 192.168.56.101
```

```
sudo nmap -p- -sV -O -sC 192.168.56.101
```



```
lavanya@kali: ~  
Session Actions Edit View Help  
(lavanya@kali)-[~]  
$ nmap -sn 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:16 +0530  
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ nmap -p- 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:17 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ nmap -sV 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:17 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -O 192.168.56.101  
[sudo] password for lavanya:  
Sorry, try again.  
[sudo] password for lavanya:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:17 +0530  
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -sC 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:18 +0530  
  
(lavanya@kali)-[~]  
$ sudo nmap -A 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:18 +0530  
  
(lavanya@kali)-[~]  
$ sudo nmap -p- -sV -O -sC 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:19 +0530  
Stats: 0:00:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 0.01% done
```

3. Windows workstation - 192.168.56.102

```
nmap -sn 192.168.56.102
```

```
nmap -p- 192.168.56.102
```

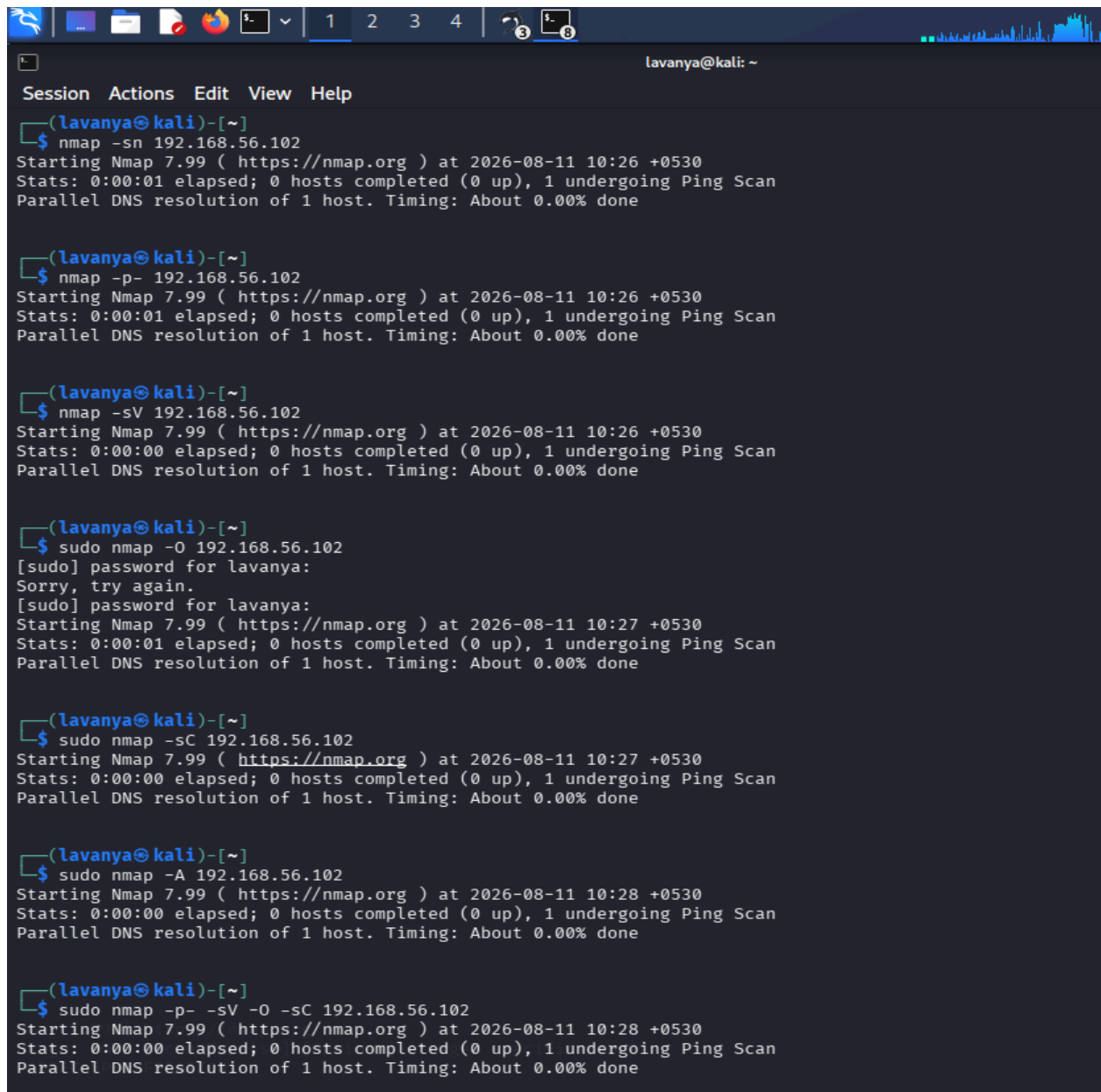
```
nmap -sV 192.168.56.102
```

```
sudo nmap -O 192.168.56.102
```

```
sudo nmap -sC 192.168.56.102
```

```
sudo nmap -A 192.168.56.102
```

```
sudo nmap -p- -sV -O -sC 192.168.56.102
```



```
(lavanya@kali)-[~]  
$ nmap -sn 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:26 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ nmap -p- 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:26 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ nmap -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:26 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -O 192.168.56.102  
[sudo] password for lavanya:  
Sorry, try again.  
[sudo] password for lavanya:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:27 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -sC 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:27 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -A 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:28 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -p- -sV -O -sC 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:28 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done
```

4. OWASP Juice Shop - 192.168.56.103

```
nmap -sn 192.168.56.103
```

```
nmap -p- 192.168.56.103
```

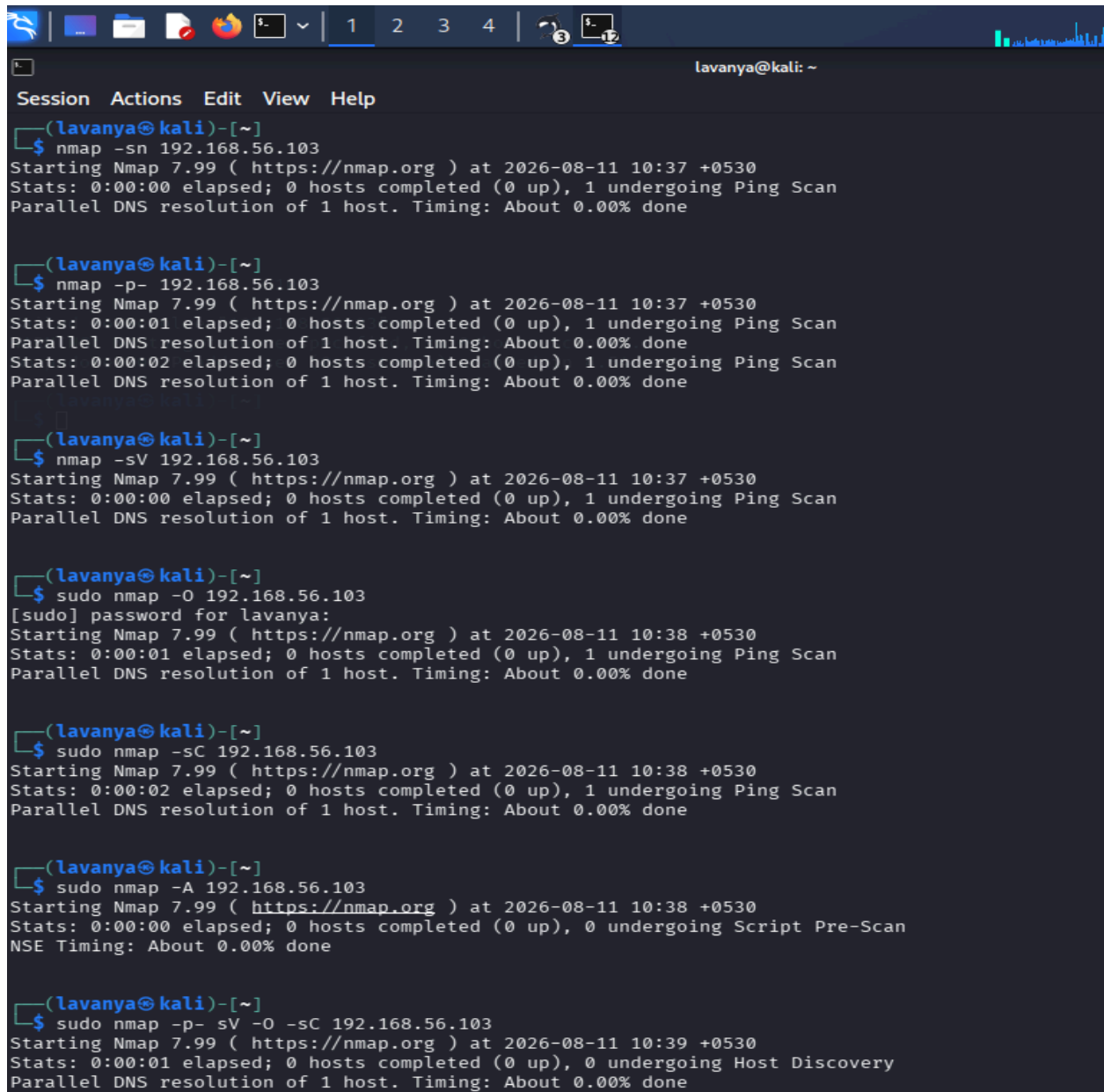
```
nmap -sV 192.168.56.103
```

```
sudo nmap -O 192.168.56.103
```

```
sudo nmap -sC 192.168.56.103
```

```
sudo nmap -A 192.168.56.103
```

```
sudo nmap -p- -sV -O -sC 192.168.56.103
```



```
(lavanya@kali)-[~]  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:37 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ nmap -p- 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:37 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ nmap -sV 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:37 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -O 192.168.56.103  
[sudo] password for lavanya:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:38 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -sC 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:38 +0530  
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -A 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:38 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan  
NSE Timing: About 0.00% done  
  
(lavanya@kali)-[~]  
$ sudo nmap -p- -sV -O -sC 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:39 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery  
Parallel DNS resolution of 1 host. Timing: About 0.00% done
```

5. Network gateway - 192.168.56.1

```
nmap -sn 192.168.56.1
```

```
nmap -p- 192.168.56.1
```

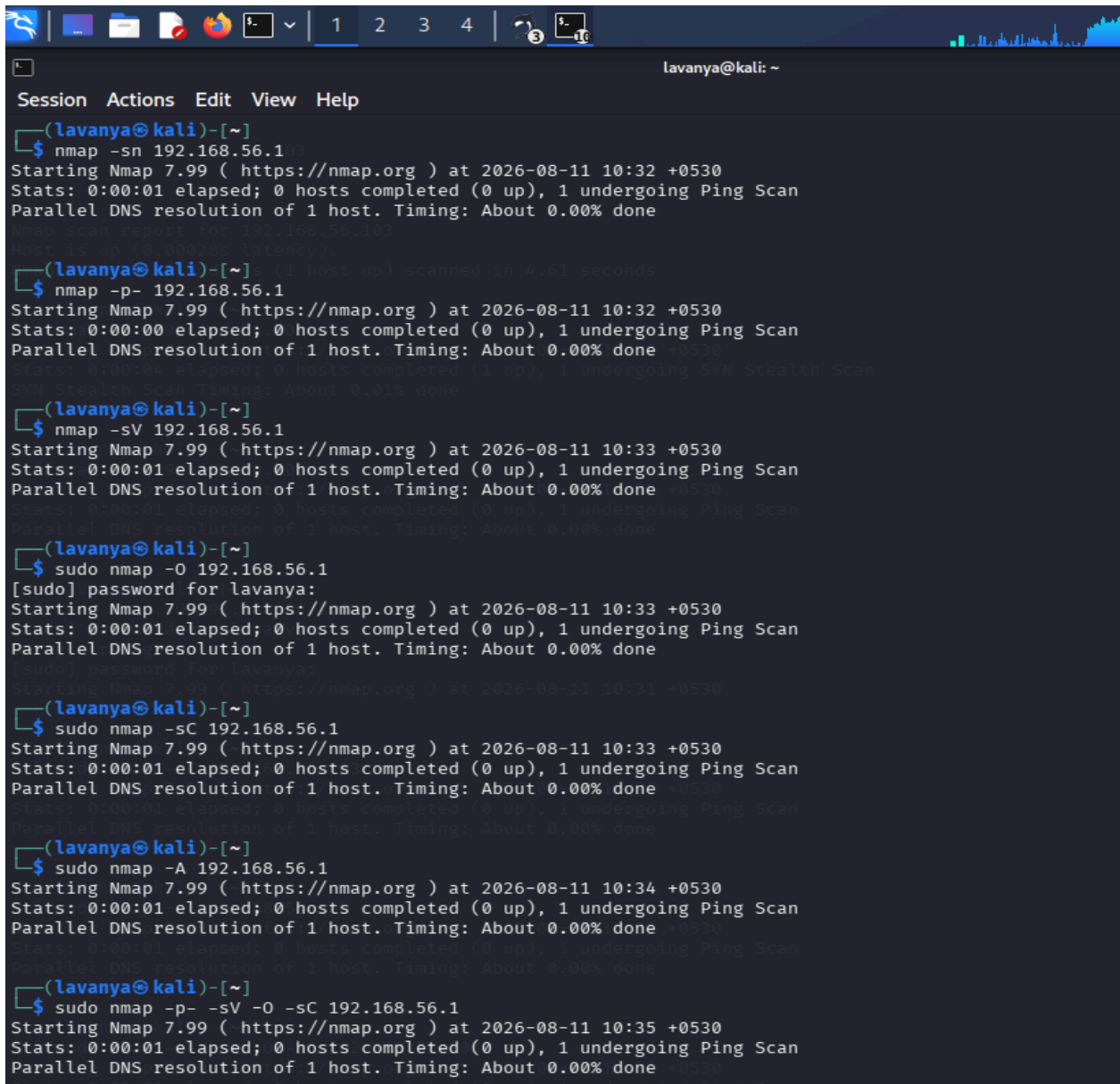
```
nmap -sV 192.168.56.1
```

```
sudo nmap -O 192.168.56.1
```

```
sudo nmap -sC 192.168.56.1
```

```
sudo nmap -A 192.168.56.1
```

```
sudo nmap -p- -sV -O -sC 192.168.56.1
```



```
lavanya@kali: ~  
Session Actions Edit View Help  
(lavanya@kali)-[~]  
$ nmap -sn 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:32 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Nmap scan report for 192.168.56.1  
Nmap: 1 host scanned in 4.61 seconds  
(lavanya@kali)-[~]  
$ nmap -p- 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:32 +0530  
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Nmap scan report for 192.168.56.1  
Nmap: 1 host scanned in 0.01 seconds  
(lavanya@kali)-[~]  
$ nmap -sV 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:33 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Nmap scan report for 192.168.56.1  
Nmap: 1 host scanned in 0.01 seconds  
(lavanya@kali)-[~]  
$ sudo nmap -O 192.168.56.1  
[sudo] password for lavanya:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:33 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Nmap scan report for 192.168.56.1  
Nmap: 1 host scanned in 0.01 seconds  
(lavanya@kali)-[~]  
$ sudo nmap -sC 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:33 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Nmap scan report for 192.168.56.1  
Nmap: 1 host scanned in 0.01 seconds  
(lavanya@kali)-[~]  
$ sudo nmap -A 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:34 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done  
Nmap scan report for 192.168.56.1  
Nmap: 1 host scanned in 0.01 seconds  
(lavanya@kali)-[~]  
$ sudo nmap -p- -sV -O -sC 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:35 +0530  
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan  
Parallel DNS resolution of 1 host. Timing: About 0.00% done
```

Code of Conduct:

I Lavanya K (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: